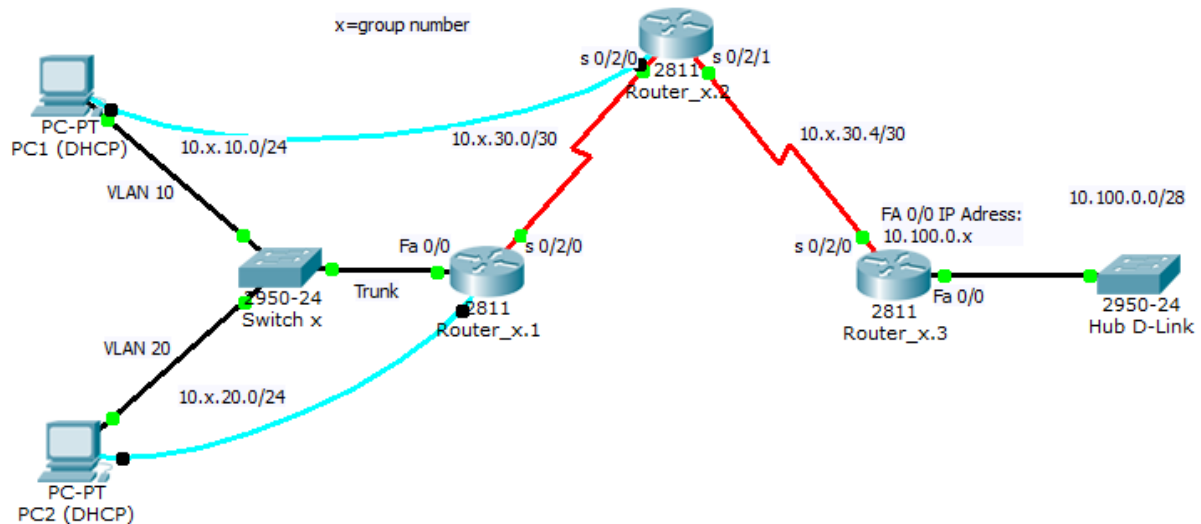


Challenge Lab: Create a network topology using real hardware - Questionary



Concerning your preparations:

Prepare the following table:

Hostname	Interface	IP-Address	Network-Address	Subnetmask	Gateway-IP
PC1	X				
PC2	X				
Router___.1	F0/0.10				X
Router___.1	F0/0.20				X
Router___.1	S0/2/0				X
Router___.2	S0/2/0				X
Router___.2	S0/2/1				X
Router___.3	S0/2/0				X
Router___.3	F0/0				x

Prepare the following table:

Hostname A	Interface A	Cable	Interface B	Hostname B
PC1				Switch
PC2				Switch
Router___.1	F0/0			Switch
Router___.2	S0/2/0 (DCE)		S0/2/0	Router___.1
Router___.2	S0/2/1 (DCE)		S0/2/0	Router___.3
Router___.3	F0/0			Hub

Concerning step 2 of the exercise: What is the difference between DCE and DTE serial connections?

Concerning step 3 of the exercise: From the routers ping the neighboring interfaces. R1 should be able to ping R2. R2 should be able to ping R3. Why can R1 not ping R3?

Concerning step 4 of the exercise: What is the difference between an access and a trunk port? Which information is needed on a trunk to assign traffic to different VLANs? Name also the IEEE standard that defines this information.

Concerning step 8 of the exercise: Determine the IP addresses assigned to PC1 and PC2 by the DHCP server.

Concerning step 10 of the exercise: Examine the routing tables of all routers. Which networks are present in the routing table and how are they connected?

Concerning step 13 of the exercise: Describe the changes that you observe on the routing table.

Concerning step 14 of the exercise: Use Wireshark to analyze RIPv2. What addresses are used as destination addresses by the routers to exchange routing information? What is the difference to RIPv1?